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6

CHEMICAL REACTIONS AND THERMAL RADIATION

6.1 Introduction

The molecular model is ideally suited to the study of chemically reacting gas flows in that the basic theory of gas phase chemical reactions is primarily concerned with processes at the molecular level (e.g. Levine and Bernstein 1987). The theory is supported by studies in experimental chemistry that are becoming increasingly concerned with measurements at the molecular level. In addition, computational methods are now being applied to the interactions between reactive molecules, but these can involve extensive calculations for each set of impact parameters. It is impossible to make such a calculation for each of the billions of collisions in a flow simulation, but it might eventually be possible to build up a database of outcomes for all impact parameters in the collisions of interest. Such a database could be accessed at each collision and would not only provide a practical approach to the chemical reaction problem, but would remove the need for the approximate molecular models that have been developed in the preceding chapters. However, the available microscopic data is fragmentary and the existing computer simulations of reacting gas flows have necessarily been based on approximate models.

The simulation models are related to the well-known collision theory for chemical reactions that is outlined in §6.2. Given the paucity of the microscopic data, the initial simulations of reacting flows (Bird 1979) employed kinetic theory to deduce the microscopic data from the continuum rate constants. This theory is presented in §6.3, while the following section deals with the extension of collision theory to termolecular collisions. This leads in §6.6 to the derivation of the ratio of forward and reverse cross-sections that produce chemical equilibrium. The combined theory is applied in §6.7 to the dissociation-recombination reaction. This section also presents a new method for these reactions that employs the Larsen–Borgnakke model for vibrational excitation in order to treat dissociation as vibrational excitation to the energy level that corresponds to dissociation.

The new method requires data only for the vibrational relaxation rate and obviates the need for explicit data on the dissociation rate. This is consistent with the basic physics of the vibration–dissociation process. The method is extended in §6.8 to the exchange reactions and again predicts rates that are consistent with the measured rates.

With regard to the application of the generalized Larsen–Borgnakke distributions of §5.5 to chemical reactions, it should be noted that the Maxwell gas versions (i.e. with $\omega = 1$) of these distributions are identical to the 'prior distributions' that are employed in theoretical chemistry (e.g. Levine and Bernstein 1987, §5.5.4). The molecular model enters into the differentiation of the set of molecules that suffer collisions from the set that constitutes the gas. The introduction of the more general results from §5.5 brings the actual distributions for inelastic collisions (the post-reaction distributions may still differ) into agreement with the prior distributions. Information theory is then largely superfluous and 'entropy maximization' procedures are no longer required for the establishment of equilibrium.

6.2 Collision theory for bimolecular reactions

Most gas phase chemical reactions may be treated as collision processes. We have seen that the VHS molecular model provides an adequate representation of real monatomic gases and that phenomenological models lead to acceptable results for gases with internal degrees of freedom. We will now investigate the extent to which a classical phenomenological approach can deal with flows that include chemical reactions and thermal radiation. A logical starting point is provided by the classical collision theory that has long been established as the simplest theory for chemical reaction rates.

The dilute gas model deals exclusively with binary collisions and is most readily extended to *bimolecular* reactions. A typical bimolecular reaction may be written schematically as

$$\mathrm{A} + \mathrm{B} \leftrightarrow \mathrm{C} + \mathrm{D}, \tag{6.1}$$

where A, B, C, and D represent separate molecular species. Note that the word 'molecule' is used in this book as a generic term and includes atoms, ions, electrons, and photons, as well as molecules. As long as the reaction takes place in a single step with no species other than the reactants present, as is implied by eqn (6.1), it has been found that the *rate equation* for the change of concentration of species A may be written

$$-\frac{\mathrm{d}n_\mathrm{A}}{\mathrm{d}t} = k_\mathrm{f}(T) n_\mathrm{A} n_\mathrm{B} - k_\mathrm{r}(T) n_\mathrm{C} n_\mathrm{D}\,. \tag{6.2}$$

The number density has been used as the measure of concentration in place of the more usual moles per unit volume, which is generally denoted by $[N_\mathrm{A}]$. The two are related by $n_\mathrm{A} = \mathcal{N}[N_\mathrm{A}]$, where $\mathcal{N}$ is Avogadro's number. The *rate coefficients* or 'constants' $k_\mathrm{f}(\mathrm{T})$ and $k_\mathrm{r}(\mathrm{T})$ relate to the forward and reverse reactions, respectively. The rate coefficients are functions of temperature and are independent of the number densities and time. A further result that originally had an empirical basis is that the rate coefficients are of the form

$$k(T) = \Lambda T^{\eta} \exp\left(-E_\mathrm{a}/kT\right), \tag{6.3}$$

where Λ and η are constants, and E_a is called the *activation energy* of the

reaction. For the special case of $\eta = 0$, this equation is called the Arrhenius equation. The formulation is essentially macroscopic, and the use of the thermodynamic temperature means that there is an implied assumption that the process does not involve a high degree of thermal non-equilibrium. This requires that the time-scale of the reaction process should be large compared with the mean collision time, thus enabling the translational and rotational velocities of the molecules to remain essentially Maxwellian.

Although the total cross-section σ_T was originally defined for elastic collisions, it did not require any modification for application to collisions that involve energy interchange between translational and fully excited internal modes. However, for chemical reactions or for the excitation of the discrete vibrational energy levels, a separate *reaction cross-section* σ_R is introduced. This is a total cross-section, but a more complete knowledge of a reaction would enable a differential cross-section to be specified as a function of the impact parameters. The ratio σ_R/σ_T of the reaction cross-section to the elastic cross-section may be regarded as the probability that an elastic collision will result in a chemical reaction. It may then be identified with the *steric factor*. However, the use of a steric factor implies that the reaction cross-section must be smaller than the elastic cross-section and, while this is generally the case, it is desirable to leave open the possibility that the reaction cross-section may exceed the elastic cross-section. If the bimolecular reaction between A and B has an activation energy E_a, it is reasonable to assume that it cannot take place unless the translational energy in the centre of mass frame of reference, together with any internal energy that is available, exceeds the activation energy. If the reaction is endothermic (i.e. it requires an energy input) the activation energy will be equal to or greater than this reaction energy. It should be noted that exothermic reactions that release energy may have a finite activation energy.

A simple but unrealistic case is to take hard sphere molecules and to assume that the reaction cross-section is zero if the translational energy along the line of centres is less than E_a and is equal to σ_R if this energy exceeds E_a. Assuming a quasi-equilibrium gas, the rate of reactive collisions of a species A molecule is given by the product of the elastic collision rate of eqn (4.75), the fraction given by eqn (4.73), and the ratio σ_R/σ_T. The forward reaction rate for the type A molecules then follows from eqn (1.39) as

$$\frac{\mathrm{d}n_A}{\mathrm{d}t} = -\frac{\sigma_R n_A n_B}{\varepsilon}\left(\frac{8kT}{\pi m_r}\right)^{1/2} \exp\left(-\frac{E_a}{kT}\right), \tag{6.4}$$

where ε is a symmetry factor that is equal to unity if $A \neq B$ and two if $A = B$.

A comparison of this equation with eqns (6.2) and (6.3) show that the collision theory result is of the expected form. However, the hard sphere model is not adequate for real gases, the reaction cross-section may not be a constant, and the internal degrees of freedom may contribute to the collision energy. The conventional collision theory result will therefore be extended to deal with a more realistic molecular model.

6.3 Reaction cross-sections for given reaction rates

Consider the case in which the collision energy E_c includes the energy in ζ_A internal degrees of freedom of the class A molecules and ζ_B internal degrees of freedom of the class B molecules. If the sum of the internal energy contributions is denoted by E_i, eqn (5.27) shows that

$$f_{E_i} \propto E_i^{\bar{\zeta}-1} \exp\{-E_i/(kT)\},$$

where $\bar{\zeta} = (\zeta_A + \zeta_B)/2$ is the average number of internal degrees of freedom that contribute to the collision energy. The distribution function for the relative translational energy E_t associated with collisions in a gas of power-law VHS molecules is given by eqn (5.13) as

$$f_{E_t} \propto E_t^{3/2-\omega_{AB}} \exp\{-E_t/(kT)\},$$

where ω_{AB} is the viscosity–temperature exponent that is specified for collisions between molecules of species A with those of species B. In this case, the full translational energy is included and the reduction that occurs when it is restricted to the energy along the line of centres will be allowed for through the energy dependence of the reaction cross-section. The distribution function for $E_c = E_t + E_i$ is given by the product of the distributions for E_i and E_t. For a fixed value of E_i, this may be written

$$f_{E_c} \propto (E_c - E_i)^{3/2-\omega_{AB}} E_i^{\bar{\zeta}-1} \exp\{-E_c/(kT)\},$$

and this distribution may be integrated over all values of E_i to give the averaged distribution function for the collision energy. The constant may be evaluated from the condition that the integrated distribution over all values of the collision energy must be unity. The final result for the distribution of E_c normalized to kT is

$$f\left(\frac{E_c}{kT}\right) = \frac{1}{\Gamma(\bar{\zeta} + 5/2 - \omega_{AB})} \left(\frac{E_c}{kT}\right)^{\bar{\zeta}+3/2-\omega_{AB}} \exp\left(-\frac{E_c}{kT}\right). \qquad (6.5)$$

The fraction of collisions with E_c greater than E_a is obtained by integrating eqn (6.5), with respect to E_c/kT, from E_a/kT to ∞. The result is

$$\frac{dN}{N} = \frac{\Gamma(\bar{\zeta}+5/2-\omega_{AB}, E_a/\{kT\})}{\Gamma(\bar{\zeta}+5/2-\omega_{AB})} \equiv Q(\bar{\zeta}+5/2-\omega_{AB}, E_a/\{kT\}). \qquad (6.6)$$

The incomplete gamma function in this equation may be reduced through eqn (B6) and this leads to particularly simple results when the first argument is an integer. For hard sphere molecules and an integer number of internal degrees of freedom, eqn (6.6) reduces to

$$\frac{dN}{N} = \exp\left(-\frac{E_a}{kT}\right)\left\{1 + \frac{E_a}{kT} + \frac{1}{2}\left(\frac{E_a}{kT}\right)^2 + \ldots + \frac{1}{(n+1)!}\left(\frac{E_a}{kT}\right)^{\bar{\zeta}+1}\right\}. \qquad (6.7)$$

For reactions in which $E_a >> kT$, the final term in the series is dominant and $s = \bar{\zeta} + 2$ is often (see, for example, Vincenti and Kruger, 1965, Chap. VII, §5) referred to as defining the number of 'square terms' that contribute energy to the reaction.

A rate coefficient of the form of eqn (6.3) is obtained (Bird 1979, 1981) if the reaction cross-section is assumed to have the form

$$\sigma_R = 0 \qquad \text{if } E_c < E_a$$

and (6.8)

$$\sigma_R = \sigma_T C_1 (E_c - E_a)^{C_2} (1 - E_a/E_c)^{\bar{\zeta}+3/2-\omega_{AB}} \qquad \text{if } E_c > E_a,$$

where C_1 and C_2 are constants. The forward reaction rate for molecules of species A is given by the combination of eqns (1.39), (4.75), (6.5), and (6.8), to give

$$\frac{dn_A}{dt} = \frac{2}{\pi^{1/2}\varepsilon} n_A n_B \sigma_{ref} \left(\frac{2kT_{ref}}{m_r}\right)^{1/2} \left(\frac{T}{T_{ref}}\right)^{1-\omega_{AB}} \frac{C_1}{\Gamma(\bar{\zeta} + 5/2 - \omega_{AB})}$$

$$\times \int_{\frac{E_a}{kT}}^{\infty} (E_c - E_a)^{C_2} (1 - E_a/E_c)^{\bar{\zeta}+3/2-\omega_{AB}} \left(\frac{E_c}{kT}\right)^{\bar{\zeta}+3/2-\omega_{AB}} \exp\left(-\frac{E_c}{kT}\right) d\left(\frac{E_c}{kT}\right).$$

If the variable in the integration is changed to $E_c - E_a$, the integral is transformed to a gamma function and the rate coefficient is

$$k(T) = \frac{2C_1\sigma_{ref}}{\pi^{1/2}\varepsilon} \left(\frac{2kT_{ref}}{m_r}\right)^{1/2} \frac{\Gamma(\bar{\zeta} + C_2 + 5/2 - \omega_{AB})}{\Gamma(\bar{\zeta} + 5/2 - \omega_{AB})} \frac{k^{C_2} T^{C_2+1-\omega_{AB}}}{T_{ref}^{1-\omega_{AB}}} \exp\left(-\frac{E_a}{kT}\right). \quad (6.9)$$

A comparison of eqn (6.9) with eqn (6.3) shows that the constants in eqn (6.8) can be written

$$C_1 = \frac{\pi^{1/2}\varepsilon\Lambda}{2\sigma_{ref}} \frac{\Gamma(\bar{\zeta} + 5/2 - \omega_{AB})}{\Gamma(\bar{\zeta} + \eta + 3/2)} \left(\frac{m_r}{2kT_{ref}}\right)^{1/2} \frac{T_{ref}^{1-\omega_{AB}}}{k^{\eta-1+\omega_{AB}}},$$

and

$$C_2 = \eta - 1 + \omega_{AB}.$$

The probability of reaction (or steric factor) at each collision with $E_c > E_a$ between power-law VHS molecules is, therefore,

$$\frac{\sigma_R}{\sigma_T} = \frac{\pi^{1/2}\varepsilon\Lambda T_{ref}^{\eta}}{2\sigma_{ref}(kT_{ref})^{\eta-1+\omega_{AB}}} \frac{\Gamma(\bar{\zeta} + 5/2 - \omega_{AB})}{\Gamma(\bar{\zeta} + \eta + 3/2)} \left(\frac{m_r}{2kT_{ref}}\right)^{1/2} \frac{(E_c - E_a)^{\eta+\bar{\zeta}+1/2}}{E_c^{\bar{\zeta}+3/2-\omega_{AB}}}. \quad (6.10)$$

This probability tends to zero as E_c tends to E_a if $\eta + \bar{\zeta} + \frac{1}{2}$ is greater than zero, but tends to infinity if this quantity is less than zero. The latter case is not physically realistic, and η should not be less than $-\frac{1}{2} - \bar{\zeta}$. Large negative values are indicative of reactions in which the internal modes contribute to the collision energy. Note also that the reaction probability tends to zero as E_c tends to infinity if η is less than $-1 + \omega_{AB}$. Dissociation reactions in air generally have values of η of the order of -1 and the reaction probability given by eqn (6.10) then has a maximum value at some value of E_c. This is the type of cross-section versus energy curve that is supported by the available evidence from molecular beam experiments (Gardiner 1969).

The contribution of the internal modes to the collision energy is an arbitrary parameter, in that the desired rate can usually be matched by a range of values of $\bar{\zeta}$, but when the above considerations are combined with the limit on the available internal energy, this range can be seen to be very limited. The resulting reaction cross-sections are generally very small in comparison with the elastic cross-sections but, for very fast reactions, they approach, but hardly ever exceed, the elastic cross-section.

The above theory for the reaction probability provides a microscopic reaction model that can reproduce the conventional rate equation in the continuum limit. However, the form of eqn (6.8) is based largely on considerations of mathematical tractability, and the model is essentially phenomenological in nature. As noted in §6.1, the ideal microscopic model would consist of complete tabulations of the differential reaction cross-sections as functions of the energy states and impact parameters. These would have to come from extensive quantum theory computations that were supported by experiment. Some microscopic data is available for particular reactions, but very little is available for reactions of engineering interest. Where comparisons can be made, the reaction cross-sections from the phenomenological theory are of the correct order. This agreement provides grounds for optimism about the validity of the results that are obtained from the application of the phenomenological model under the highly non-equilibrium conditions that are generally encountered in rarefied gas flows.

6.4 Extension to termolecular reactions

A termolecular reaction involves three molecules and is related to ternary or three-body collisions. This is an important reaction in high-temperature air in that it is the reverse or recombination reaction in the dissociation of molecular oxygen and nitrogen. A typical dissociation–recombination reaction can be represented as

$$AB + T \leftrightarrow A + B + T. \tag{6.11}$$

Here, AB represents the dissociating molecule, A and B are atoms (or at least one molecule if AB is polyatomic), and T is the 'third body' atom or molecule. There is an energy E_d required for the dissociation which is an *endothermic* reaction. A similar amount of energy is associated with the *exothermic* recombination, and it might appear that the reverse reaction in eqn (6.13) could be a bimolecular reaction rather than a *termolecular* reaction involving three molecules. However, it is easily seen that eqn (2.1) for the momentum and eqn (2.2) for the energy in a binary collision cannot be simultaneously satisfied in the presence of heat release. Three-body or ternary collisions are therefore required for recombination. The rate of formation of either species A or species B in the forward reaction can be written

$$dn_A/dt = k_f n_{AB} n_T, \tag{6.12}$$

while the loss of species A in the reverse reaction is

$$dn_A/dt = k_r n_A n_B n_T. \tag{6.13}$$

Note that the product of the rate constant for a termolecular reaction and a number density has the same dimensions as the rate constant in a bimolecular reaction.

The collision theory can be extended to ternary reactions by assigning a 'lifetime' to a binary collision between two molecules, and then considering triple collisions as a second binary collision between the combination of the first two molecules with a third molecule. Alternatively, the recombination can be regarded as the combination of two binary collisions. The first of these forms an orbiting pair that is stabilized by the second collision with the third body, as long as the second collision occurs within a sufficiently small elapsed time. The two approaches are effectively equivalent and, whatever the dependence of the lifetime on the relative speed and on the translational energy in the first binary collision, the probability of the recombination will be directly proportional to the number density n_T of the third body species. If the recombination rate corresponding to the reverse reaction in eqn (6.1) is specified by a rate equation of the form of eqn (6.3), the recombination probability at each binary collision between a species A molecule and a species B molecule can be obtained in a similar manner to eqn (6.10). If the activation energy is taken as zero, the contribution from the internal degrees of freedom is also zero, and the recombination probability that is associated with the binary collision is

$$\frac{\sigma_R}{\sigma_T} = \frac{\pi^{1/2} n_T \varepsilon \Lambda T_{ref}^{\eta}}{2\sigma_{ref}} \frac{\Gamma(5/2 - \omega_{AB})}{\Gamma(\eta + 3/2)} \left(\frac{m_r}{2kT_{ref}}\right)^{1/2} \left(\frac{E_c}{kT_{ref}}\right)^{\eta - 1 + \omega_{AB}}. \tag{6.14}$$

This probability tends to infinity as E_c tends to zero unless $\eta - 1 + \omega_{AB}$ is equal to or greater than zero. The value of ω_{AB} is close to 0.75, so that ω should not be less than about 0.25. This is more restrictive in practice than the corresponding constraint for the bimolecular reaction.

6.5 Chemical equilibrium

Chemical equilibrium requires a balance between the forward and reverse reactions such that the concentrations of all species remain constant with time. Therefore, from eqn (6.2) for a bimolecular reaction,

$$K_{eq} \equiv \frac{k_f}{k_r} = \frac{n_C n_D}{n_A n_B} \tag{6.15}$$

where K_{eq} is the *equilibrium constant*. Similarly, the equilibrium constant for the termolecular dissociation–recombination reaction of eqn (6.11) follows from eqns (6.12) and (6.13) as

$$K_{eq} \equiv \frac{k_f}{k_r} = \frac{n_A n_B}{n_{AB}}. \tag{6.16}$$

For the dissociation of *symmetrical* or *homonuclear* diatomic molecules in which species A is the same as species B, this becomes

$$K_{eq} = \frac{k_f}{k_r} = \frac{n_A^2}{n_{AA}}. \tag{6.17}$$

The *degree of dissociation* α may now be defined as the mass fraction of dissociated gas. Therefore,

$$\alpha = \frac{n_A m_A}{\rho} = \frac{n_A}{n_A + 2n_{AA}}. \tag{6.18}$$

The combination of eqns (6.17) and (6.18) shows that

$$\frac{\alpha^2}{1-\alpha} = \frac{m_A}{2\rho}\frac{k_f}{k_r}. \tag{6.19}$$

Equilibrium statistical mechanics relates the numbers of the various species in a system to the *partition functions*. For the bimolecular reaction,

$$\frac{N_C N_D}{N_A N_B} = \frac{Q^C Q^D}{Q^A Q^B} \exp\left(-\frac{E_a}{kT}\right), \tag{6.20}$$

where Q is the partition function for the species that is indicated by the superscript, and E_a should be interpreted here as the difference between the forward and backward activation energies. The number N is related to the volume V of the system by

$$N = nV,$$

and the combination of eqns (6.15) and (6.20) enables the law of mass action for this reaction to be written

$$K_{eq} \equiv \frac{k_f}{k_r} = \frac{n_C n_D}{n_A n_B} = \frac{Q^C Q^D}{Q^A Q^B} \exp\left(-\frac{E_a}{kT}\right). \tag{6.21}$$

In the case of the termolecular dissociation–recombination reaction, the system volume does not cancel and the corresponding equation is

$$K_{eq} \equiv \frac{k_f}{k_r} = \frac{n_A n_B}{n_{AB}} = \frac{Q^A Q^B}{V Q^{AB}} \exp\left(-\frac{E_a}{kT}\right). \tag{6.22}$$

Equation (6.19) for the symmetrical molecule can then be written

$$\frac{\alpha^2}{1-\alpha} = \frac{m_A}{2\rho V} \frac{(Q^A)^2}{Q^{AA}} \exp\left(-\frac{E_d}{kT}\right). \tag{6.23}$$

Each partition function may be written as the product of the separate translational, rotational, vibrational, and electronic partition functions for that species, i.e.

$$Q = Q_{tr} Q_{rot} Q_v Q_{el}. \tag{6.24}$$

For the reactions that are encountered in high-temperature air and over the temperature range of interest, approximate expressions are available for the partition functions. First, the translational partition function is

$$Q_{tr} = V(2\pi m k T/h^2)^{3/2}. \tag{6.25}$$

Note that the equilibrium constants are independent of the system volume when this function is applied to either bimolecular or termolecular reactions. The rotational partition function is generally written

$$Q_{rot} = T/(\varepsilon' \Theta_r) \tag{6.26}$$

where ε' is a symmetry factor equal to 2 for homonuclear molecules and 1 for heteronuclear molecules. The harmonic oscillator model leads to a vibrational partition function of the form

$$Q_v = 1/\{1 - \exp(-\Theta_v/T)\}. \tag{6.27}$$

Finally, the electronic partition function is summed over the j levels, i.e.

$$Q_{el} = \sum_{i=0}^{j} g_i \exp(-\Theta_{el,i}/T), \tag{6.28}$$

where g_i are the degeneracies (i.e. the number of states in the level), and the ground state energy $\Theta_{el,0}$ is set to zero.

The partition functions have been used to calculate the equilibrium composition of reacting gases, and some results from their application to high-temperature air are shown in Figs 1.9 and 1.10. They can also be used to calculate the backward rate coefficient from the forward coefficient, or to test the consistency of listed forward and backward coefficients.

Reaction cross-section data at the microscopic level should be consistent with the principle of detailed balancing. Because chemical reactions involve the internal as well as the translational states, the formulation of the reciprocity principle for a reacting gas would be a formidable task, but the data can be employed in a way that leads to chemical equilibrium.

6.6 The equilibrium collision theory

Continuum studies of flows that involve chemical reactions almost always employ rate equations in the form of eqn (6.3) for both the forward and reverse reactions. However, the simple form of eqn (6.3) does not match the mathematical complexity of the partition functions, and each set of rate equations is consistent with chemical equilibrium only over a narrow range of temperature. This temperature range will, in general, be different for each reaction and, for a complex gas with many simultaneous reactions, the correct equilibrium state will not be attained at any temperature. The same error arises also in particle based calculations if the theory of §6.3 and §6.4 is used to determine the reaction cross-sections from both the forward and reverse rate equations.

The above difficulty can be overcome if the theory is used to calculate the reaction cross-sections only for the forward reaction, and the reaction cross-sections for the reverse reactions are related to those for the forward reactions through the partition functions. The temperature in the partition functions must be the local macroscopic temperature.

The reactions almost invariably require an activation energy in only one direction, and the reaction in this direction is chosen to be the forward reaction. The cross-section for the reverse reaction can then be set to a fraction F of the elastic cross-section and, for a bimolecular collision, this fraction is a function only of the macroscopic temperature. In the case of a termolecular reverse reaction, it is also proportional to the number density of the molecules that constitute the third bodies in the collisions. In both cases the fraction is independent of the collision energy, i.e.

$$\sigma_R / \sigma_T = F \tag{6.29}$$

for the bimolecular reaction, and

$$\sigma_R / \sigma_T = n_T F \tag{6.30}$$

for the termolecular reaction.

The change with time of the number density of species C in the reverse bimolecular reaction is

$$\mathrm{d}n_C / \mathrm{d}t = n_C \nu_{CD} F.$$

The equilibrium collision rate of eqn (4.75) may be used for ν_{CD} and the resulting equation combined with eqns (1.39) and (6.2) to give

$$k_r = (2/\varepsilon)\pi^{-\frac{1}{2}}\sigma_{ref}(T/T_{ref})^{1-\omega}(2kT_{ref}/m_r)^{\frac{1}{2}}F, \tag{6.31}$$

where the values of σ_{ref}, T_{ref}, and ω are the values appropriate to the collision pair for the reverse reaction, and ε. It is easily seen that this equation applies also to the termolecular reverse reaction as long as F is defined by eqn (6.30), rather than (6.29). The forward reaction rate is in the form of eqn (6.3) and may be written

$$k_f = \Lambda_f T^{\eta_f} \exp\{-E_a/(kT)\}. \tag{6.32}$$

Equations (6.31) and (6.32) for the rate coefficients may be substituted into eqn (6.21) to give, for the bimolecular reaction,

$$\frac{\Lambda_f T^{\eta_f} \exp\{-E_a/(kT)\}}{(2/\varepsilon)\pi^{-\frac{1}{2}}\sigma_{ref}(T/T_{ref})^{1-\omega}(2kT_{ref}/m_r)^{\frac{1}{2}}F} = \frac{Q^C Q^D}{Q^A Q^B}\exp\left(-\frac{E_a}{kT}\right). \quad (6.33)$$

The exponential term disappears and, using eqn (6.29),

$$\frac{\sigma_R}{\sigma_T} = \frac{\Lambda_f T^{\eta_f}}{(2/\varepsilon)\pi^{-\frac{1}{2}}\sigma_{ref}(T/T_{ref})^{1-\omega}(2kT_{ref}/m_r)^{\frac{1}{2}}}\frac{Q^A Q^B}{Q^C Q^D}. \quad (6.34)$$

A similar procedure for the termolecular reverse reaction gives

$$\frac{\sigma_R}{\sigma_T} = \frac{\Lambda_f T^{\eta_f}}{(2/\varepsilon)\pi^{-\frac{1}{2}}\sigma_{ref}(T/T_{ref})^{1-\omega}(2kT_{ref}/m_r)^{\frac{1}{2}}}\frac{n_T V Q^{AB}}{Q^A Q^B}. \quad (6.35)$$

As expected, the reaction probabilities are functions of the macroscopic temperature. While this is less physically realistic than the dependence on E_c for the forward reactions, the probabilities lead to exact equilibrium.

The theory may be extended to cover the comparatively rare reactions that have a finite activation energy $E_{a,r}$ for the reverse reaction. The energy in the exponential term on the right-hand side of eqn (6.33) becomes the difference between the forward and reverse activation energies. In addition, the collision rate must be reduced by multiplying it by the fraction of collisions that have energies in excess of $E_{a,r}$. This fraction in an equilibrium gas at temperature T is given by eqn (6.6), and the modified expression for the bimolecular reaction is

$$\frac{\sigma_R}{\sigma_T} = \frac{\pi^{\frac{1}{2}}\Lambda_f T^{\eta_f}\Gamma(\bar{\zeta}+5/2-\omega)\exp(-E_{a,r}/\{kT\})(Q^A Q^B)/(Q^C Q^D)}{(2/\varepsilon)\sigma_{ref}(T/T_{ref})^{1-\omega}(2kT_{ref}/m_r)^{\frac{1}{2}}\Gamma(\bar{\zeta}+5/2-\omega, E_{a,r}/\{kT\})} \quad (6.36)$$

where $\bar{\zeta}$ is now the average number of internal degrees of freedom that contribute to $E_{a,r}$. This reaction probability is applied only to those reactions that have energy greater than $E_{a,r}$. The same factors can be applied to eqn (6.35) for the termolecular reactions.

6.7 The dissociation–recombination reaction

The preceding theory may be illustrated through its application to the dissociation and recombination reaction. For example, for oxygen

$$O_2 + T \leftrightarrow 2O + T. \quad (6.37)$$

Typical rates for the reaction that can be applied for all collision partners or 'third bodies' are given by Gupta et al. (1990) as

$$k_f = 6.0\times10^{-12}\,T^{-1}\exp\{-8.2\times10^{-19}/(kT)\}$$

and (6.38)

$$k_r = 8.3\times10^{-45}\,T^{-\frac{1}{2}}.$$

The units are such that the rate coefficient is in $m^3 molecule^{-1} s^{-1}$ for the binary reaction and $m^6 molecule^{-2} s^{-1}$ for the ternary reaction.

The application of the rates of eqn (6.38) implicitly assumes that the equilibrium constant is

$$K_{eq} = 7.23\times10^{32}\, T^{-\frac{1}{2}} \exp(-59400/T). \tag{6.39}$$

On the other hand, the equilibrium constant from statistical mechanics is given by eqn (6.22) as

$$K_{eq} = \frac{(Q^{O})^2}{V Q^{O_2}} \exp\left(-\frac{E_a}{kT}\right).$$

The substitution into this equation of the approximate partition functions of eqns (6.25) to (6.28), and the incorporation of the appropriate data for oxygen from Appendix A gives

$$\begin{aligned} K_{eq} = {} & 1.785\times10^{28} T^{\frac{1}{2}} \{1-\exp(-2270/T)\} \exp(-59400/T) \\ & \times\{5+3\exp(-228.9/T)+\exp(-325.9/T)+5\exp(-22830/T) \\ & +\exp(-48621/T)\}^2/\{3+2\exp(-11393/T)+\exp(-18985/T)\}. \end{aligned} \tag{6.40}$$

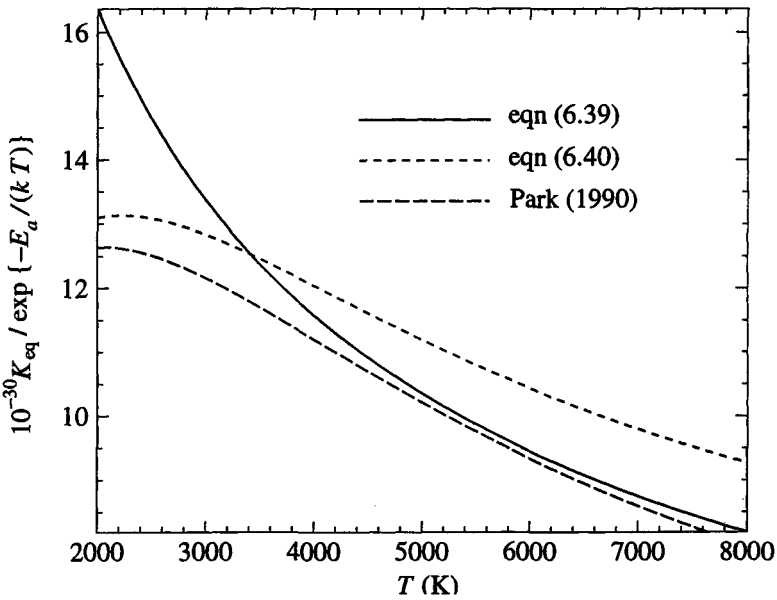


Fig. 6.1 A comparison of the equilibrium constant from typical reaction rates with that from statistical mechanics.

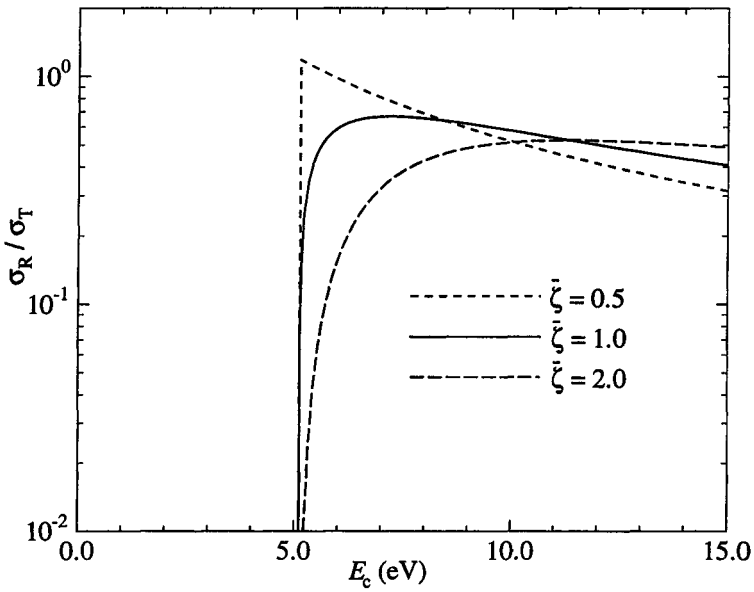


Fig. 6.2 Reaction probabilities for the dissociation of oxygen.

The number of electronic states has been truncated through the omission of those that make only small contributions at the temperatures of interest. In addition, the vibrational function is based on the harmonic oscillator model and neglects rotational-vibrational coupling. These effects have been included in a study by Jaffe (1987) and the resulting partition functions and equilibrium constants have been tabulated by Park (1990).

The comparison in Fig. 6.1 of the equilibrium constants from eqns (6.39) and (6.40) demonstrates that they are in agreement only at a temperature of 3500 K. However, the more accurate equilibrium constant of Park (1990) shows that the rates of eqn (6.38) lead very nearly to equilibrium at higher temperatures. The errors in the approximate partition functions are less serious for nitrogen.

The forward rate of eqn (6.38) may be converted to collision energy dependent steric factors by eqn (6.10). In order to avoid the singularity in this equation when $E_c = E_a$, the average number of internal degrees of freedom that contribute to E_c must not be less than 0.5. Figure 6.2 shows the reaction probabilities as a function of the collision energies for this value, and for several higher values. The reaction cross-section can exceed the elastic cross-section for $\bar{\zeta} = 0.5$. However, a larger contribution from the internal degrees of freedom is to be expected and these lead to steric factors that are more realistic from the physical point of view.

The incorporation of the reverse reaction rate of eqn (6.38) into eqn (6.14) leads to an infinite reaction probability as $E_c \to 0$. This may be avoided by setting the temperature index η to $1 - \omega_{AB}$, so that the power of E_c is zero and the ratio of the reaction cross-section to the elastic cross-section is a constant. The change in η should be compensated for by changing the constant Λ so that k_r is unchanged at the temperature of interest. When this is done for a temperature of 3500 K, and noting that the third-body number density is here equal to the overall number density, the result is

$$\sigma_R / \sigma_T = 0.9 \times 10^{-4} n/n_0, \tag{6.41}$$

where n_0 is the number density under standard conditions. This procedure is consistent with the 'equilibrium collision theory' that was presented in the preceding section. If $\eta = 1 - \omega_{AB}$ in eqn (6.14) and the reverse rate equation that appears in this equation is replaced by the forward rate equation through eqn (6.22), eqn (6.14) becomes identical with eqn (6.35). However, the cross-section ratio or steric factor of eqn (6.41) leads only approximately to equilibrium and it is preferable to dispense completely with the reverse rate equation and to base the reaction cross-sections for the reverse reactions on the equilibrium collision theory. This theory applies to bimolecular as well as termolecular reverse reactions and can also deal with reverse reactions that have a finite activation energy.

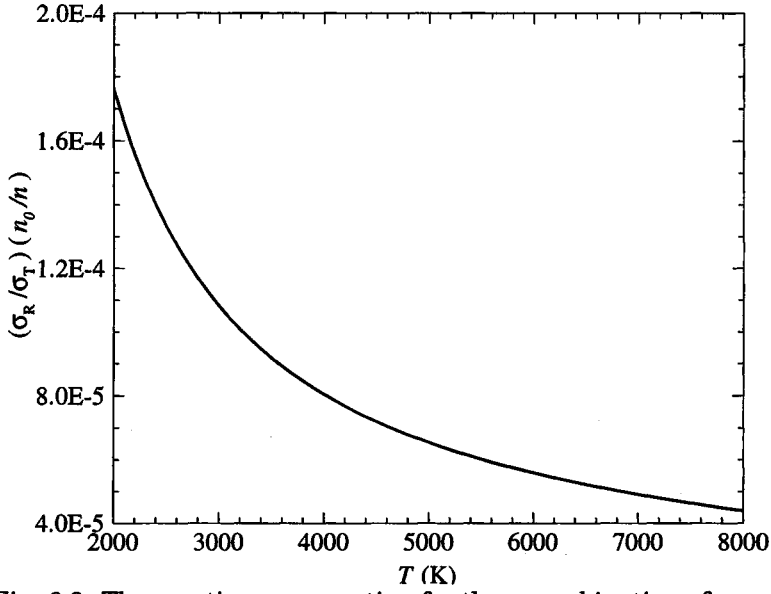


Fig. 6.3 The reaction cross-section for the recombination of oxygen.

The evaluation of eqn (6.35) for the recombination of oxygen using the forward reaction rate of eqn (6.38) and the approximate partition functions of eqns (6.25)–(6.28) leads to the results shown in Fig. 6.3. The ratio σ_R/σ_T is the probability of recombination in any binary collision between two O atoms. This is independent of the collision energy, but depends on the gas temperature. The accuracy could be improved through the use of the tabulated values (Park 1990) of the partition functions. The dependence on the number density reflects the fact that the recombination reaction is termolecular.

Although it can match the experimentally determined reaction rates and can lead to equilibrium at all temperatures, the preceding theory is essentially phenomenological in nature and it would be preferable to have a theory that is more closely related to the physics. Dissociation occurs when the vibrational energy of the molecule reaches the value that corresponds to the dissociation energy. It is therefore desirable to combine the theories for dissociation and vibrational excitation. The rate at which the individual molecules in an equilibrium gas attain the dissociation energy E_d in the vibrational mode may be calculated from the Larsen–Borgnakke theory. While this theory is also phenomenological, its parameters are inferred from the measured vibrational relaxation rate.

Consider a collision that is inelastic with respect to the vibrational excitation of the diatomic molecule that is subject to dissociation. The probability that the vibrational energy exceeds the dissociation energy may be written down directly from eqn (5.47). Ξ_b is set equal to the sum of the translational contribution and the total number of post-collision internal modes, so that the energy $E_a + E_b$ is equal to the total collision energy E_c. If molecule 1 dissociates in the collision, there are no rotational or vibrational degrees of freedom associated with the resulting atoms, so that all the internal energy in this molecule goes into breaking the interatomic bond. First consider the case in which the third body molecule (subscript 2) is also diatomic. The collision energy is sufficiently high that the vibrational mode as well as the rotational mode must be included in Ξ_b, which becomes $9/2 - \omega$. The required probability is, therefore,

$$P' = (1 - E_d/E_c)^{9/2-\omega}. \tag{6.42}$$

This probability applies only to collisions in which E_c exceeds E_d, and the probability of this is given by the integration of eqn (6.5). The overall probability that $\varepsilon_{v,1}$ exceeds E_d is obtained by including the probability of eqn (6.42) in this integration, i.e.

$$P_{\varepsilon_{v,1}>E_d} = \frac{1}{\Gamma(\bar{\zeta}+5/2-\omega)} \int_{E_d/(kT)}^{\infty} \left(\frac{E_c}{kT}\right)^{\bar{\zeta}+3/2-\omega} \exp\left(-\frac{E_c}{kT}\right) P' \, \mathrm{d}\left(\frac{E_c}{kT}\right). \tag{6.43}$$

The average number of internal degrees of freedom in eqn (6.43) refers to the pre-collision state of the two molecules. If it is assumed that the vibrational modes are fully excited, the average number is four, and eqn (6.43) becomes

$$P_{\varepsilon_{v,1}>E_d} = \frac{1}{\Gamma(13/2-\omega)} \int_{E_d/(kT)}^{\infty} \left(\frac{E_c}{kT}\right)\left(\frac{E_c-E_d}{kT}\right)^{9/2-\omega} \exp\left(-\frac{E_c}{kT}\right) d\left(\frac{E_c}{kT}\right).$$

This may be evaluated through a simple change of variable to give

$$P_{\varepsilon_{v,1}>E_d} = \left(1 + \frac{E_d}{(11/2-\omega)kT}\right)\exp\left(-\frac{E_d}{kT}\right). \tag{6.44}$$

In the application of the Larsen-Borgnakke theory, the sampling from the distributions is applied to only a fraction of the collisions. A separate fraction is applied to each mode, and the inverses of these fractions define the rotational and vibrational relaxation collision numbers. The Landau-Teller theory for vibrational relaxation predicts vibrational collision numbers of the order of unity at temperatures of the order of 20,000 K, and it is reasonable to assume that energy redistribution occurs in all collisions that are sufficiently energetic to be included in the integration of eqn (6.43). This essentially assumes a steric factor of unity in the collisions that meet the energy criterion, and the expression on the right-hand side of eqn (6.44) becomes the fraction of collisions that lead to dissociation.

The number of dissociations per unit volume per unit time is then given by the product of this fraction with the number of collisions between the relevant molecular species per unit volume per unit time in the gas. The equilibrium collision rate given by eqns (4.74) and (1.39) as

$$N_{12} = 2\pi^{-1/2}\sigma_{ref} n_1 n_2 (T/T_{ref})^{1-\omega}(2kT_{ref}/m_r)^{1/2}.$$

The symmetry factor has been omitted from this rate because, when both molecule 1 and molecule 2 are of the same species, *either* molecule could meet the requirement for dissociation. This can be combined with eqns (6.12) and (6.44) to give the following expression for the rate coefficient

$$k_f = \frac{2\sigma_{ref}}{\pi^{1/2}}\left(\frac{T}{T_{ref}}\right)^{1-\omega}\left(\frac{2kT_{ref}}{m_r}\right)^{1/2}\left(1 + \frac{E_d}{(11/2-\omega)kT}\right)\exp\left(-\frac{E_d}{kT}\right). \tag{6.45}$$

Most reactions are sufficiently slow that the use of the equilibrium collision rate does not lead to significant error. A correction may be required for very fast reactions, and this point has been discussed by Baras and Nicolis (1990).

In the case of the dissociation of a diatomic molecule in a collision with an atom, all the internal energy must be in molecule 1 and Ξ_b is $5/2-\omega$. Also, the mean number of degrees of freedom of the pre-collision molecules is two, and the final result is

$$k_f = \frac{2\sigma_{ref}}{\pi^{1/2}}\left(\frac{T}{T_{ref}}\right)^{1-\omega}\left(\frac{2kT_{ref}}{m_r}\right)^{1/2}\left(1 + \frac{E_d}{(7/2-\omega)kT}\right)\exp\left(-\frac{E_d}{kT}\right). \tag{6.46}$$

This theory is essentially an improvement on the 'available energy' theory for dissociation that was proposed by Fowler and Guggenheim (1952), and applied by Hansen (1965). While the original theory required an arbitrary choice for the number of internal degrees of freedom that contribute to the required dissociation energy, the division of energy is now defined by the Larsen–Borgnakke distribution function and there are no disposable parameters. However, it should be noted that these equations assume a vibrational collision number of unity at the temperature that corresponds to dissociation. They could be made more general through the inclusion of a finite Z_v in the denominator.

The prediction of eqn (6.45) for the oxygen rate coefficient is compared in Fig. 6.4 with the forward coefficient from eqn (6.38). The corresponding rate given by Gupta et al. (1990) for the dissociation of nitrogen is

$$k_f = 3.2\times10^{-13}\, T^{-\frac{1}{2}} \exp\{-1.56\times10^{-18}/(kT)\}, \tag{6.47}$$

and this is also compared with the prediction of eqn (6.45). The rate coefficients of eqns (6.38) and (6.47) are based on curve fits of eqn (6.3) to experimental data. It is generally accepted that the data for the forward dissociation rates is subject to at least a factor of three uncertainty and it is clear that the predicted dissociation rate is of the correct magnitude.

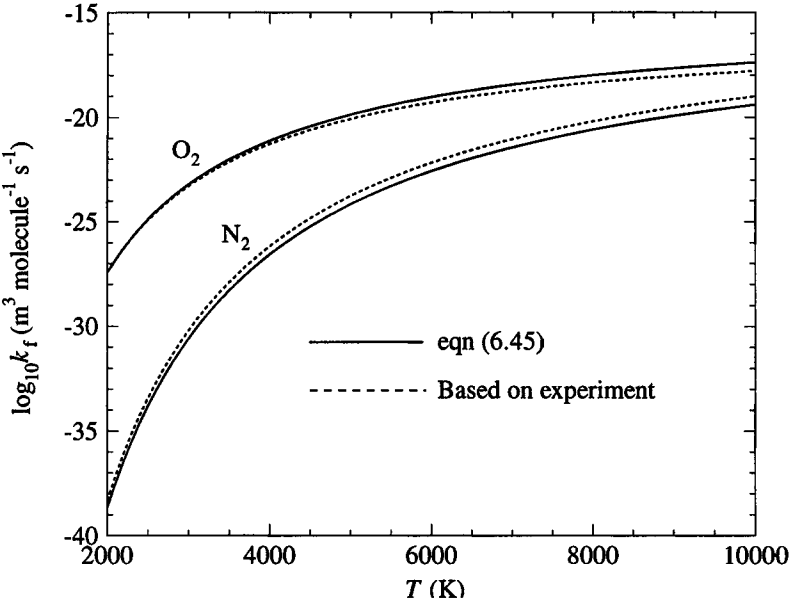


Fig. 6.4 Rate coefficients for the dissociation of oxygen and nitrogen.

Bird (1994) compares the theoretical rate coefficients from eqns (6.45) and (6.46) with a range of measured coefficients for the dissociation of oxygen and nitrogen. The theoretical values generally lie within, or just below, the range of experimental results. The agreement tends to be better at temperatures above 10,000 K, and the variation with temperature is in close agreement with experimentally based coefficients that have $\eta = -\frac{1}{2}$. The experimental rate coefficients for dissociation by an atom are generally higher that those for dissociation by a molecule, but the factor is generally larger than that predicted by the theory.

It must be kept in mind that a completely classical theory such as this cannot account for the quantum effects that exist in real gases. While it is not surprising that these effects appear to be relatively small for the dissociation of moderately heavy molecules such as oxygen and nitrogen, light gases and ionization reactions should be approached with caution.

This 'exact available energy' theory for dissociation is compatible with the 'equilibrium collision' theory for recombination that was developed in §6.5. If the mean number of internal degrees per molecule in the pre-collision pair $\bar{\zeta}$ is retained explicitly, eqns (6.45) and (6.46) can be written in a form that applies to all third-body molecules. This is

$$k_f = \frac{2\sigma_{ref}}{\pi^{1/2}} \left(\frac{T}{T_{ref}}\right)^{1-\omega} \left(\frac{2kT_{ref}}{m_r}\right)^{1/2} \left(1 + \frac{E_d}{(\bar{\zeta} + 3/2 - \omega)kT}\right) \exp\left(-\frac{E_d}{kT}\right). \quad (6.48)$$

and it may be combined with the version of eqn (6.35) for like molecules, to give the probability of recombination as

$$\frac{\sigma_R}{\sigma_T} = 2C \left(1 + \frac{E_d}{(\bar{\zeta} + 3/2 - \omega)kT}\right) \frac{n_T V Q^{A_2}}{(Q^A)^2}. \quad (6.49)$$

The constant C, which is close to unity, is the ratio of the mean product of relative velocity and collision cross-section in the dissociating collisions to the same quantity in the recombining collisions.

6.8 The exchange and ionization reactions

There are several reactions that change the identity of the dissociated atoms in air under high-temperature conditions. These are

$$NO + O \leftrightarrow O_2 + N$$

with a suggested (Gupta et al. 1990) rate coefficient

$$k_f = 5.3\times10^{-21}\, T \exp\{-2.72\times10^{-19}/(kT)\},$$

and

$$N_2 + O \leftrightarrow NO + N \quad (6.50)$$

with a suggested forward rate coefficient

$$k_f = 1.12\times10^{-16} \exp\{-5.17\times10^{-19}/(kT)\}.$$

The general Larsen–Borgnakke result of eqn (5.47) may also be applied to these reactions. If the rotational mode of the diatomic molecule is included in the second group, the parameter Ξ_b is 7/2–ω, so that

$$P' = (1 - E_a/E_c)^{7/2-\omega}.$$

Note that E_a in this equation is the activation energy of the exchange reaction, rather than the energy in the first group of modes, as in eqn (5.47). The mean pre-collision number of internal degrees of freedom is two and the integration of eqn (6.49) leads to the remarkably simple result

$$P_{\varepsilon_{r,1}>E_a} = \exp\{-E_a/(kT)\}. \tag{6.51}$$

Because the activation energy is now much smaller than the dissociation energies, a finite vibrational relaxation number Z_v must be included in the denominator, i.e.

$$k_f = \frac{2\sigma_{ref}}{\pi^{1/2} Z_v}\left(\frac{T}{T_{ref}}\right)^{1-\omega}\left(\frac{2kT_{ref}}{m_r}\right)^{1/2}\exp\left(-\frac{E_a}{kT}\right). \tag{6.52}$$

The vibrational relaxation collision number τ_v is the product of the collision rate and the relaxation time. The Landau-Teller theory predicts that the relaxation time is related to temperature raised to the power of minus one third. Experimental data is available for the relaxation time in common gases at temperatures of the order of 5,000 K, and this confirms the Landau–Teller prediction. Millikan and White (1963) have shown that the measured data can be fitted to an expression of the form

$$p\tau_v = C\exp(C_2 T^{-1/3}),$$

where C and C_2 are constants. This equation can be combined with eqn (1.28) for the pressure and with either eqn (4.64) or eqn (4.74) for the collision rate to obtain an expression for the vibrational collision number as a function of the temperature. This is, with C_1 as another constant,

$$Z_v = (C_1/T^{\omega})\exp(C_2 T^{-1/3}). \tag{6.53}$$

The rate coefficients predicted by eqn (6.52), with the vibrational collision number of eqn (6.53) evaluated at the temperature corresponding to the activation energy of the forward reaction, is compared in Fig. 6.5 with the experimentally based coefficients of eqn (6.56). In the absence of specific data for NO, the data for N_2 in Table A6 is applied to both gases. Given the uncertainty in the vibrational collision number, the agreement with the continuum rate data is again satisfactory.

It is not clear that the rotational energy should play any part in the exchange reactions, and the possible consequences of this uncertainty can be investigated by ignoring the rotational mode. The parameter Ξ_b in the general Larsen–Borgnakke distribution is then 5/2–ω and

$$P' = (1 - E_a/E_c)^{5/2-\omega}.$$

The mean precollision number of internal degrees of freedom is now one and the integration of eqn (6.43) again leads to

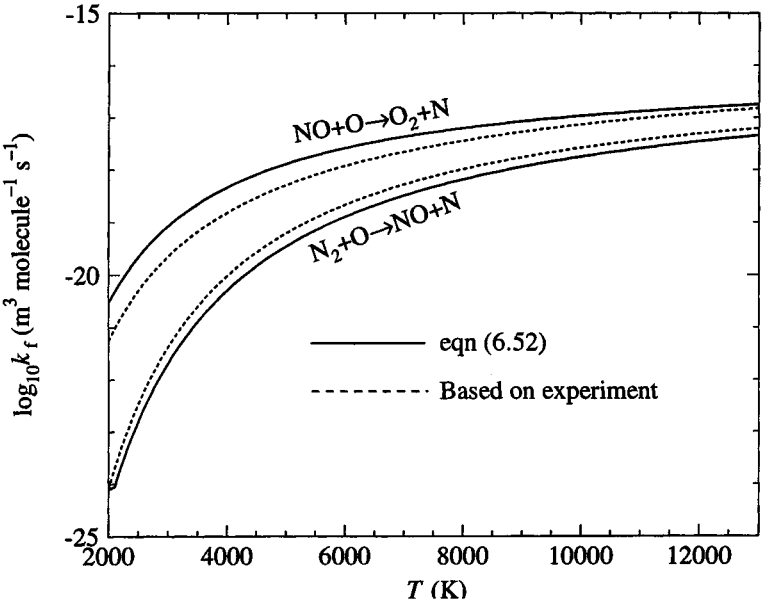


Fig. 6.5 Rate coefficients for the exchange reactions.

$$P_{\varepsilon_{v,1}>E_a} = \exp\{-E_a/(kT)\}. \tag{6.54}$$

The increased probability of energy going into the vibrational mode due to the neglect of the post-collision rotation is therefore exactly balanced by the omission of the pre-collision rotation from the calculation of the probability that E_c exceeds E_a. This is consistent with the analytical justification of the serial application of the Larsen–Borgnakke method in §5.5.

Eqn (6.54) is therefore a general result for collisions in which the number of internal modes does not change. It defines the fraction of collisions in which the post-collision energy in a single mode will exceed the energy E_a. This fraction is compared in Fig. 6.6 with the fraction of collisions in which the total energy E_c exceeds E_a. The latter fraction is given by eqn (6.6) and this function has been plotted for $\omega = 0.75$ and a range of values for the mean number of internal degrees of freedom that contribute to E_c. The effect of ω on this function has already been shown in Fig. 4.6.

Ionization reactions involve the electronic states and, not only are the electronic levels widely and unevenly spaced, but the transitions between them are subject to restrictive selection rules. This means that simple classical theories are unlikely to be successful in predicting the rate coefficients of ionization reactions. Nevertheless, the available energy theory should provide an upper limit for these rates.

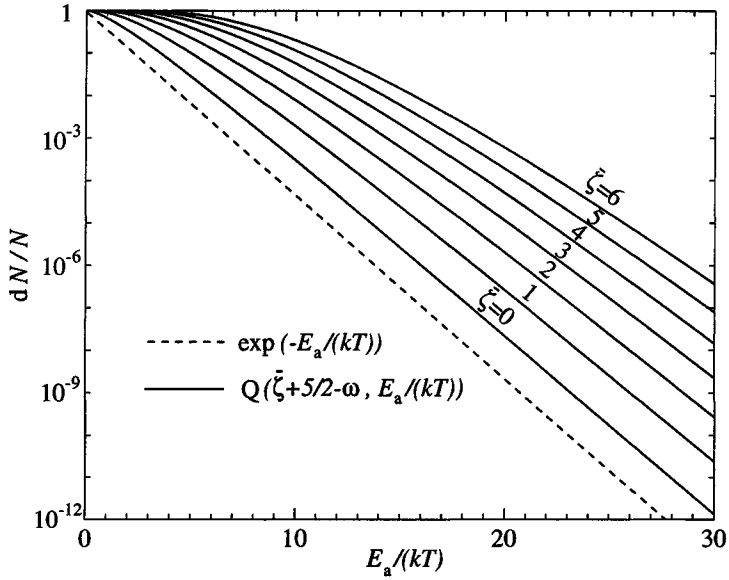


Fig. 6.6 The functions that determine the fraction of collisions that satisfy the various energy constraints.

Consider the ionization of molecular nitrogen in an impact with another nitrogen molecule, i.e.

$$N_2 + N_2 \leftrightarrow N_2^+ + e + N_2 .$$

for which a typical rate constant is

$$k_f = 2.5\times10^{-22}\, T^{½} \exp\{2.5\times10^{-18}/(kT)\}. \tag{6.55}$$

The average molecular energy associated with the electronic excitation is

$$\varepsilon_{el} = \Big(k \sum_{i=0}^{j} g_i \Theta_{el,i} \exp(-\Theta_{el,i}/T) \Big) / Q_{el} , \tag{6.56}$$

and the contribution from each level is kT when $T = \Theta_{el}$ for that level. The probability that the ionization energy is attained in the electronic mode of one molecule is approximately that given by eqn (6.51) with the activation energy set to the ionization energy E_i. The rate coefficient is then similar to eqn (6.52), or

$$k_f = \frac{2\sigma_{ref}}{\pi^{½} Z_i} \left(\frac{T}{T_{ref}}\right)^{1-\omega} \left(\frac{2kT_{ref}}{m_r}\right)^{½} \exp\left(-\frac{E_i}{kT}\right).$$

A comparison of this equation with eqn (6.55) leads to a steric factor of approximately 0.0001. This is essentially the ratio of the ionization to the elastic cross-section.

The important reaction in the initial ionization in air is the associative–ionization reaction, rather than the direct impact ionization. This is

$$N + O \leftrightarrow NO^+ + e,$$

with a typical rate constant of

$$k_f = 1.5\times10^{-20}\, T^{1/2} \exp\{4.5\times10^{-19}/(kT)\}. \tag{6.57}$$

The application of the general Larsen–Borgnakke distribution to this case requires a numerical integration of eqn (6.43), and the fraction of 'available energy' collisions that leads to reactions is less than one in a hundred.

6.9 Classical model for rotational radiation

Bound–bound radiation may be emitted as a consequence of transitions between rotational states. This is significant only for heteronuclear molecules that have a permanent electric dipole moment, and this type of radiation need not be considered when dealing with symmetric diatomic molecules such as oxygen and nitrogen. Rieger (1974) has compared the quantum and classical models for rotational emission and has shown that the classical model can produce useful results.

The classical model regards the molecule as a rotating dipole and the radiated power is matched by a reduction in the rotational energy of the molecule. The magnitude of the angular velocity is related to the rotational energy E_r by

$$\omega = (2E_r/I)^{1/2}, \tag{6.58}$$

where I is the moment of inertia. The wavelength of the radiation is

$$\lambda = 2\pi c/\omega = (2\pi^2 c^2 I/E_r)^{1/2} \tag{6.59}$$

where c is the speed of light. After a time interval Δt, the angular velocity decays to

$$\omega' = \omega/(1 + \delta\omega^2\Delta t)^{1/2},$$

where

$$\delta = \frac{\mu_e^2}{3\pi c^3 I \varepsilon_0}. \tag{6.60}$$

Here μ_e is the electric dipole moment and ε_0 is the free space dielectric constant.

The corresponding change in the wavelength of the radiation is

$$\lambda' = \lambda(1 + \kappa\Delta T/\lambda^2)^{1/2},$$

where

$$\kappa = \frac{4\pi\mu_e^2}{3\varepsilon_0 cI}. \tag{6.61}$$

The radiation energy emitted over this time interval is

$$E_{\text{rad}} = \tfrac{1}{2} I(\omega^2 - \omega'^2) = 2\pi^2 c^2 I(1/\lambda^2 - 1/\lambda'^2). \tag{6.62}$$

The HF molecule may be used to illustrate the magnitudes of the quantities in these equations. The moment of inertia is 1.34×10^{-47} kg m^2 and the electric dipole moment is 6.07×10^{-30} C m. A typical angular velocity at temperature T is $(kT/I)^{\frac{1}{2}}$ and, at a temperature of 2,000 K, this is 4.5×10^{13} s^{-1}. The wavelength associated with this frequency 4.2×10^{-5} m. This indicates that rotational radiation generally falls in the infra-red region of the electromagnetic spectrum.

6.10 Bound–bound thermal radiation

Bound–bound radiation occurs as a result of radiative transitions between the quantized energy levels of atoms and molecules. These radiative events occur spontaneously and are not associated directly with inter-molecular collisions. The transitions in atoms are between the electronic levels but, in the case of molecules, radiative transitions involve the rotational, vibrational, and electronic levels. Because the levels designate the energy in the particular mode, a transition between a particular pair of levels involves a fixed energy e_p of the photon. This energy is related to the frequency by

$$e_p = h\nu, \tag{6.63}$$

so that bound–bound transitions give rise to emission and absorption *lines*.

The probability of the spontaneous emission is expressed through the *Einstein coefficient* A_{nm} which is defined such that the probability per unit time that a particle at energy state n will undergo a radiative transition to state m, with the radiant energy in the element of solid angle $d\Omega$, is

$$A_{nm}\, d\Omega.$$

The number of $n \rightarrow m$ transitions per unit time per unit volume is therefore

$$-(dn/dt)_{n\rightarrow m} = 4\pi n_n A_{nm} = n_n/\tau_{nm}, \tag{6.64}$$

where τ_{nm} is the *mean radiative lifetime* for the transition. Einstein coefficients are also defined for the absorption and for the induced emission of radiation. However, it may be shown (e.g. Vincenti and Kruger 1965) that the coefficients are related by equilibrium constraints, and only one of the three is independent.

Typical magnitudes of the mean radiative lifetime and the wavelength of the radiation are shown in Table 6.1. The mean radiative lifetimes are several orders of magnitude greater than the mean collision time at standard density, but become comparable with this time at low densities.

Table 6.1 Molecular band systems in air.

Band	Transition	τ (s)	λ (m)
N_2,1+	$3 \rightarrow 2$	1.1×10^{-5}	1.06×10^{-6}
N_2,2+	$5 \rightarrow 3$	2.7×10^{-8}	0.34×10^{-6}
O_2,S-R	$5 \rightarrow 1$	8.2×10^{-9}	0.20×10^{-6}
NO,β	$3 \rightarrow 1$	6.7×10^{-7}	0.22×10^{-6}
NO,γ	$2 \rightarrow 1$	1.2×10^{-7}	0.23×10^{-6}

There are then insufficient collisions to maintain the equilibrium fraction of molecules at the higher electronic levels, and the radiation is said to be 'collision limited'.

In an equilibrium gas, the fraction of molecules with electronic states in level i is given by the Boltzmann distribution of eqn (5.63). If this equation applies, the gas is said to be in 'local thermal equilibrium'. For the non-equilibrium case, the Larsen–Borgnakke method can be used to select the electronic states, as described in §5.6. This requires data on the ratio of the excitation to the elastic cross-section.

In the cases of the inelastic collisions and chemical reactions, it was possible to formulate a phenomenological model at the particle level such that the description becomes identical with that given by the continuum model when the density is sufficiently high for this model to be valid. This does not appear to be possible for thermal radiation.

References

BARAS, F and NICOLIS, G. (1990). Microscopic simulations of exothermic chemical systems, In *Microscopic simulations of complex flows*, (ed. M. Mareschal), pp 339–352, Plenum Press, New York.

BIRD, G.A. (1979). Simulation of multi-dimensional and chemically reacting flows, In *Rarefied gas dynamics*, (ed. R. Campargue), pp 365–388, CEA, Paris.

BIRD, G.A. (1981). Monte Carlo simulation in an engineering context, *Progr. Astro. Aero.*, **74**: Part I, 239–255.

BIRD, G.A. (1994). A new chemical reaction model for DSMC calculations, *Prog. in Astro. and Aero.* **159**, 185-196.

FOWLER, R.H. and GUGGENHEIM, E.A. (1952). *Statistical thermodynamics*, Cambridge University Press.

GARDINER, W.C. (1969). *Rates and mechanisms of chemical reactions*, Benjamin, New York.

GUPTA, R.N., YOS, J.M., THOMPSON, R.A., and LEE, K-P (1990). A review of reaction rates and thermodynamic and transport properties for an 11-species air model for chemical and thermal nonequilibrium calculations to 30 000 K, NASA Reference Publication 1232.

JAFFE, R.L. (1987). The calculation of high-temperature equilibrium and non-equilibrium specific heat data for N_2, O_2, and NO, Am. Inst. Aero. and Astro. Paper AIAA-87-1633.

HANSEN, C.F. (1965).Estimates for collision-induced dissociation rates. *AIAA J.* **3**, 61–66.

LEVINE, R.D. and BERNSTEIN, R.B. (1987). *Molecular reaction dynamics and chemical reactivity*, Oxford University Press.

MILLIKAN, R.C. and WHITE, D.R. (1963). Systematics of vibrational relaxation. *J. Chem. Phys.* **39**, 3209–3213.

PARK, CHUL (1990). *Nonequilibrium hypersonic aerothermodynamics*, Wiley, New York.

REIGER, T.J. (1974). Pure rotational emission from diatomic molecules in the quantum and classical limits, *J. Quant. Spectrosc. Radiat. Transfer*, **14**, 59–67.

VINCENTI, W.G. and KRUGER, C.H. (1965). *Introduction to physical gas dynamics*, Wiley, New York.